

ENRIQUE GARCIA RIVERA

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EDUCATION

Washington University in St. Louis

Expected: December 2026

B.S. Mechanical Engineering

GPA: 3.64 / 4.00

Relevant Coursework: Modeling, Simulation & Control (MATLAB/Simulink), Numerical Methods & Matrix Algebra, Fluid Mechanics, Heat Transfer, Thermodynamics, Solid Mechanics, Vibrations, Electrical & Electronic Circuits

SKILLS

- **Software and Tools:** MATLAB/Simulink, Python (proficient), C++ (working knowledge), SolidWorks, ANSYS Mechanical, PX4 autopilot, Gazebo simulation, XFLR5, Git/version control, embedded systems (Teensy, Arduino)
- **Engineering Competencies:** Mechanical design and rapid prototyping, structural and aerodynamic analysis, sensor instrumentation and fusion, avionics systems integration, experimental testing and validation.

EXPERIENCE

WashU VTOL – Avionics & Systems Integration Lead, St. Louis, MO

Sep 2025 – Present

- Designed and integrated avionics architecture for a prototype eVTOL aircraft including Pixhawk 6C flight controller, GPS, IMU, telemetry radios, and power distribution systems.
- Configured PX4 autopilot and navigation sensors to support reliable communication between onboard systems and ground monitoring tools.
- Conducted staged subsystem validation through bench testing and HIL simulation, collecting and analyzing flight data to verify system stability and sensor performance.
- Diagnosed and resolved integration issues across sensors, computing hardware, and communication systems, documenting architecture and test procedures for repeatable troubleshooting.

MIRO Lab, UMKC – Undergraduate Researcher (Grant Funded), Kansas City, MO

May – Aug 2025

- Built an embedded sensing platform synchronizing IMU and EMG sensor streams at 1 kHz on a Teensy 4.0, applying physical intuition about sensor behavior to design calibration and noise characterization procedures.
- Characterized sensor noise, bias drift, and timing jitter across motion capture and arm movement trials to evaluate measurement reliability.
- Collected and prepared IMU and EMG datasets across controlled motion trials to support Extended Kalman Filter development for multi-sensor state estimation.
- Developed Python scripts for signal processing, filtering, and statistical analysis to validate sensor fusion accuracy across experimental conditions.

WashU Design Build Fly – Aerodynamics & Payload Engineer, St. Louis, MO

Sep 2024 – Jan 2026

- Evaluated aircraft wing geometries using XFLR5 to analyze lift, drag, and stability tradeoffs, applying aerodynamic principles to support configuration selection.
- Performed structural finite element analysis in ANSYS Mechanical to verify wing spar load capacity and structural safety margins under expected flight loading.
- Designed and fabricated mechanical components including payload release mechanisms using SolidWorks CAD modeling and additive manufacturing.
- Supported flight test operations across multiple sessions, assisting with aircraft setup and logging flight data for post-flight review.

Washington University McKelvey School of Engineering – Course Assistant, St. Louis, MO

Aug – Dec 2025

- Supported MEMS 2000 Electrical & Electronic Circuits through laboratory sessions, instrumentation troubleshooting, and circuit debugging with students.

TECHNICAL PROJECTS

- **Hyperion Orbital Physics Simulator (C++):** Implemented a six-DOF orbital dynamics simulator with J2 perturbation modeling to deepen understanding of spacecraft motion and numerical integration techniques.
- **Flight Navigation EKF System (Embedded C++):** Built a Kalman filter fusing GPS, barometer, and IMU data to explore state estimation concepts and their application to real-time autonomous navigation.
- **Sentinel Prototype (C++ and Python):** Designed and assembled a robotic sensing system with embedded closed-loop control and multi-sensor feedback to gain practical experience in autonomous tracking applications.

INVOLVEMENT

Society of Hispanic Professional Engineers (SHPE)

Sep 2023 – Present

American Society of Mechanical Engineers (ASME)

Aug 2024 – Present

Kessler Scholars Collaborative, Washington University

Jun 2023 – Present